EE3042 Energy Conversions (3+0) 6

Spring 2022

Lectures: Wednesday, 10:30-12:20, Thursday, 09:30-12:20

Instructor: Dr. Mustafa Onat
Marmara University, Faculty of Engineering, Office: M4-237, Maltepe
monat@marmara.edu.tr

Course Outline: Energy technology and resources: Fossil fuels, nuclear, solar, and other types of energy. Three phase systems and magnetic circuits. Transformers: Ideal and physical models and equivalent circuit, and transformer testing. Electromechanical energy conversion. Efficiency and process performance. Sensors and actuators: Relays, stepper and positioning systems, switched reluctance machines, synchronous reluctance machines, direct current (DC) machines. Symmetrical alternating current (AC) synchronous machines. Symmetrical AC induction (asynchronous) machines.

Objectives:

1. Calculate/measure single-phase and three-phase real and reactive power flow.
2. Analyze and design simple magnetic devices relating mechanical dimensions to magnetic quantities.
3. Analyze the performance of practical single-phase and three-phase transformers.
4. Discuss electric-and magnetic-field interactions in electromechanical devices and machines.
5. Analyze the steady-state performance of practical synchronous, induction, and DC machines.

Outcomes: This course provides an introduction to energy conversion techniques, with an emphasis on rotating magnetic field-based machinery. Through lecture and out-of-class assignments, students are provided learning experiences that enable them to:

1. Analyze three-phase networks under balanced conditions.
2. Analyze basic magnetic circuits, such as inductors and transformers.
3. Understand the concept of electromagnetic force production, analyze basic electromechanical actuators.
4. Understand fundamental concepts associated with rotating electrical machinery, such as traveling magneto-motive force (mmf) waves and pole numbers.
5. Analyze rotating induction and synchronous machines to determine performance indicators such as torque and efficiency.

Text Book: Electric Machinery Fundamentals, Stephen J. Chapman, 5th Edition, McGRAW-HILL, 2012. A. E. Fitzgerald, Electric Machinery, Sixth Edition, Electric Machinery and Transformers, Bhag S. Guru, Hüseyin R. Hızıroğlu, 3rd Edition

Honor Code: All work done on the exams will be done on your own and pledged. Homework concepts and approaches may be discussed with other students, but the work will be done by the individual. Group or copied solutions are not permitted and *negative full grade* will be given for such work.

Attendance: Classroom attendance is mandatory. Pop-quizzes will be given at unscheduled and random times. No make-ups will be arranged for pop-quizzes.

Examinations: Exams will be closed book and in-class.

Grades:

For regular students

Project	15%
Quiz	5%
Midterm	40%

For repeat students

Project	20%
Quiz	0%
Midterm	40%

Tentative Schedule

Week	Date (Please arrange this column)	Subjects	Chapters	Study Questions
1	23.02.2024	Introduction to Electric Machinery Principles	1	Questions:1.1 - 1.19 Problems: 1.3, 1.5, 1.6, 1.8, 1.10
2	2.03.2022	Magnetic Circuits	1	Problems: 1.12, 1.13,1.15, 1.17, 1.19
3	9.03.2022	DC Linear Machines	1	Problems: 1.21, 1.22, 1.23
4	16.03.2022	DC Machinery Fundamentals	7	Questions: 7.1, 7.2, 7.6, 7.7, 7.9, 7.10, 7.12, 7.13, 7.16
5	23.03.2022	DC Machinery Fundamentals	7	Problems: 7.1, 7.2, 7.3, 7.4 7.6, 7.8, 7.9, 7.10
6	30.03.2022	Transformers	2	Questions: 2.1 - 2.11
7	6.04.2022	Transformers	2	Problems:2.1, 2.4 - 2.9, 2.11 – 2.15, 2.17-2.20, 2.22-2.24
	13.04.2022	**Midterm**		
8	20.04.2022	DC Motors and Generators	8	
9	27.04.2022	DC Motors and Generators	8	
10	4.05.2022	AC Machinery Fundamentals	3	
11	11.05.2022	Synchronous Generators	4	
12	18.05.2022	Synchronous Generators	4-5	
13	25.05.2022	Synchronous Motors	5	
14	1.06.2022	Synchronous Motors	6	
15	8.06.2022	Induction Motors	6	
16	15.06.2022	Induction Motors	6	
17		**Final Exam**		